

MID TERM

**CS304 - Object Oriented Programming**

**Question No: 1 ( Marks: 1 ) - Please choose one**

What problem(s) may occur when we copy objects without using deep copy constructor?

- ▶ Dangling pointer
- ▶ Memory Leakage
- ▶ All of the given
- ▶ System crash

**Question No: 2 ( Marks: 1 ) - Please choose one**

Suppose that the Test class does not have an overloaded assignment operator. What happens when an assignment  $a=b$ ; is given for two Test objects a and b?

- ▶ The automatic assignment operator is used
- ▶ The copy constructor is used
- ▶ Compiler error
- ▶ Run-time error

**Question No: 3 ( Marks: 1 ) - Please choose one**

a static member function can be called, even when a class is not \_\_\_\_\_.

- ▶ Declared
- ▶ Define
- ▶ Instantiated
- ▶ Called

**Question No: 4 ( Marks: 1 ) - Please choose one**

Identify which of the following overloaded operator function's declaration is appropriate for the given call?

Rational\_number\_1 + 2.325

Where Rational\_number\_1 is an object of user defined class Rational\_number.

- ▶ Rational\_number operator+( Rational\_number & obj);
- ▶ Rational\_number operator+(double& obj); (Page 145)
- ▶ Rational\_number operator+(Rational\_number &obj, double& num);
- ▶ operator+(double& obj);

**MID TERM**

**Question No: 5 ( Marks: 1 ) - Please choose one**

\_\_\_\_\_ provide the facility to access the data member.

- ▶ **accesser function** (Page 68)
- ▶ private function
- ▶ inline function
- ▶ None of the given

**Question No: 6 ( Marks: 1 ) - Please choose one**

Constant objects cannot change their state,

- ▶ **True** (Page 105)
- ▶ False

**Question No: 7 ( Marks: 1 ) - Please choose one**

The \_\_\_\_\_ relationship indicates that an object contains other objects.

- ▶ **None of given** (Page 53)
- ▶ 'has-a'
- ▶ 'is-a'
- ▶ 'be-

**Question No: 8 ( Marks: 1 ) - Please choose one**

Which one of the following features of OOP is used to derive a class from another?

- ▶ Encapsulation
- ▶ Polymorphism
- ▶ Data hiding
- ▶ **Inheritance** (Page 25)

**Question No: 9 ( Marks: 1 ) - Please choose**

one \_\_\_\_\_ is a relationship

- ▶ **Inheritance** (Page 25)
- ▶ Polymorphism
- ▶ abstraction
- ▶ encapsulation

**Question No: 10 ( Marks: 1 ) - Please choose one**

\_\_\_\_\_ satisfy the condition of polymorphism

- ▶ Carbon
- ▶ Diamond
- ▶ Coal
- ▶ **all of the given** (Page 56)

**MID TERM**

**Question No: 11 ( Marks: 1 ) - Please choose one**

A generalization-specialization relation between classes are implemented using

- ▶ data hiding
- ▶ friend classes
- ▶ encapsulation
- ▶ inheritance (Page 49)

**Question No: 12 ( Marks: 1 ) - Please choose one**

The >= operator can be overloaded.

- ▶ True (Page 140)
- ▶ False

**Question No: 13 ( Marks: 1 ) - Please choose one**

In order to free the memory occupied by the object, we use -----

- ▶ Constructor
- ▶ Destructor (Page 92)
- ▶ Shallow Copy
- ▶ Deep Copy

**Question No: 14 ( Marks: 1 ) - Please choose one**

Which of the following is not an example of multiple inheritances?-----

- ▶ Mermaid
- ▶ Woman (Lecture No.5)
- ▶ None of the given
- ▶ Amphibious Vehicle

**Question No: 15 ( Marks: 1 ) - Please choose one**

Static variable can be initialized more than once.

- ▶ True
- ▶ False

**Question No: 16 ( Marks: 1 ) - Please choose one**

A generic class showing all the common attributes and a behavior of other classes represents a very important feature in oop called -----

- ▶ Inheritance (Page 26)
- ▶ Encapsulation
- ▶ Polymorphism
- ▶ Abstraction

MID TERM

**Question No: 17 ( Marks: 2 )**

Which type of copy is used by the default assignment operator "=" when applied to objects. Also describe the reason.

**Question No: 18 ( Marks: 2 )**

Consider the following code segment:

```
class User {  
    char * name;  
    const int id;  
public:  
    User(){  
        Id=123; // It gives an error  
    }  
};
```

So how we can assign a value to constant id?

**Question No: 19 ( Marks: 3 )**

Give c++ code to overload assignment operator for string class.

**Question No: 20 ( Marks: 3 )**

Consider the code given below explain what kind of association exists between class A and class B. Justify your answer as well.

```
class A{  
  
private:  
    int a,b,c;  
  
public:  
    ....  
};  
  
class B{  
  
private:  
    int d,e,f;  
    A obj1;  
public:  
    ....  
};
```

MID TERM

---

**Question No: 21 ( Marks: 5 )**

How we can use the concept of overloading to convert a type according to our own requirements? Give one example.

**Question No: 22 ( Marks: 5 )**

Consider the following scenario,

In a University a Course consists of 45 Lectures, 6 Assignments, 1 Quiz and 1 GMDB, similarly each Lecture has corresponding 1 general MDB; you have to model this scenario as an **Object Model** clearly showing type of associations (**simple association, composition or aggregation**) between these Objects.